

BIOLOGY X

Questions Distribution

According to the Chapters.

**This Distribution is carried out According to
NEW BOOK OF BIOLOGY.**

COMPILED BY Dr. Areeba Kamran

QUESTIONS		Years
UNIT:1	GASEOUS EXCHANGE	
Section "B"		
3. What is lungs Cancer?		2025
2. Why is smoking dangerous for health? And how is it related with respiratory disorders?		2024
6. How is the asthma patient characterized?		2022
Section "C"		
15. Draw a neat and labelled diagram of human respiratory system		2025
14. Draw a neat and labelled diagram of respiratory system of human.		2023
SOME IMPORTANT QUESTIONS FROM NEW BOOK		
Section "B"		
1. Why the stomata generally open during day-time?		
2. Which parts of the plant intake CO_2 and give out O_2 , take in O_2 oxygen and give out CO_2 during day-time?		
3. Why do we have to breathe through nostrils rather than oral cavity?		
4. Differentiate between breathing, gaseous exchange and respiration.		
5. Why do we deep breath during or immediately after exercise?		
6. What is "oxygen debt" ?		
7. Distinguish between inspiration and expiration.		
8. How the asthma is characterized?		
9. Name five animals which use their body surface for gaseous exchange.		
Section "C"		
1. What measures would you take to avoid respiratory disorders?		

2. Discuss human respiratory system with the help of suitable illustrations.
3. Prove with the help of experiment that CO_2 is released during respiration.
4. Explain the process of ventilation in man.
5. Why smoking is dangerous? How it is related with respiratory disorders?

UNIT:2**HOMEOSTASIS****Section "B"**

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| 5. What are three important aspects of homeostasis? Define any two of them. | 2025 |
| 6. Write three characteristics of Hydrophytes. | |
| 10. Draw a neat and labelled diagram of L.S of human skin. | |
| 12. What are xerophytes plants? | 2024 |
| 5. Define excretion in animals. Name the excretory organ of any three animals. | 2023 |
| 7. Why is skin considered as excretory organ? | 2022 |

Section "C"

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| 15. Draw a neat and labelled diagram of human Kidney or Nephron. | 2024 |
| 18. Write down names of disorders of kidney. Explain any two from them. | 2023 |
| 15. Describe the urinary system of man with the help of diagram. | 2022 |

SOME IMPORTANT QUESTIONS FROM NEW BOOK**Section "B"**

1. Why homeostasis is required?
2. Why plants remove liquid water instead of water vapours?
3. How plant survive in saltish water?
4. What type of structures are present in dermis to perform different functions?
5. Draw a neat and labelled diagram of nephron.
6. What is kidney stone?
7. How human skin maintains temperature in cold conditions?
8. When dialysis is required?
9. Why filtration at Para tabular capillaries called ultra filtration?

Section "C"

1. How skin works as thermoregulatory organ?

2. Describe the structure of nephron within the L.S of kidney.
3. Describe the network of blood vessels in nephron and their functions.
4. Describe different disorders of kidneys and their treatment.

UNIT:3**COORDINATION AND CONTROL****Section "B"**

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| 2. What is the role of Islet's of Langerhans cell? | 2025 |
| 3. What is Epilepsy? | 2024 |
| 4. Write three differences between chemical and nervous co-ordination. | 2023 |
| 6. Draw a neat and labelled diagram of human eye. | |

Section "C"

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| 14. Describe in detail the structure of Human Brain. | 2025 |
| 16. Describe the structure and functions of human Ear or Eye. | 2024 |
| 18. Define Endocrine gland. Describe enzymes secreted from pituitary and thyroid gland. | |
| 17. Define endocrine system. Write down name of three endocrine glands with one secretion of each. | 2023 |
| 14. Draw neat and labelled diagram of human ear. | 2022 |

SOME IMPORTANT QUESTIONS FROM NEW BOOK**Section "B"**

1. What do you mean by feedback system?
2. Why the nervous coordination is faster than chemical coordination?
3. Which of the two coordination types is better and why?
4. How reflex action works by a reflex-arc?
5. Why driving license is not issued to a colour blind person?
6. Why thyroid gland swell up and give the name of the disease?
7. What is "emergency hormone" and why it is named so?
8. What are the possible reasons of paralysis?
9. What is Epilepsy?
10. Differentiate between chemical and Nervous Coordination.

Section "C"

1. Write a note on ear.
2. Why does the intelligence level of persons differ from each other, even though the components of brain are same?

3. What is endocrine system? Discuss the gland involved in regulation of blood glucose and how?
4. What is diabetes mellitus? Explain some ways of its management.

UNIT:4**SUPPORT AND MOVEMENT****Section "B"**

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| 13. What are the three main types of skeleton? Describe any one of them with example. | 2025 |
| 4. Write three differences between bone and cartilage. | 2024 |
| 5. Describe any three functions of joint. | |
| 10. Write down three functions of skeleton. | 2023 |
| 5. What is rickets? Write its causes. | 2022 |
| 10. Draw neat and labelled diagram of hinge OR ball and socket joint. | |

SOME IMPORTANT QUESTIONS FROM NEW BOOK**Section "B"**

1. What are the main functions of skeleton?
2. Distinguish between bone and cartilage.
3. Locate the cartilages in different organs.
4. Enlist the name and number of bones present in hind limb.
5. Locate the immovable joints in the human body.
6. What is ricket? Give its causes.
7. Distinguish between ligament and tendon.
8. Locate hinge joint in the body of human
9. What do you mean by bicep and tricep muscles?

Section "C"

1. Prove that the skeletal system is a dynamic system
2. Describe the flexion and extension in human arm
3. Explain disorders of bones and their causes.

UNIT:5**REPRODUCTION****Section "B"**

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| 8. Draw a neat and labelled diagram of T.S of angiospermic flower OR structure of Bean seed. | 2025 |
| 9. Differentiate between sexual and asexual reproduction. | |
| 11. Draw a neat and labelled diagram of Ovule. | 2024 |
| 12. Draw a neat and labelled diagram of ovule. | 2023 |

3. Draw neat and labelled diagram of Ovule.	2022
12. Distinguish between sexual and asexual reproduction.	
Section "C"	
15. —(OR)— Draw a neat and labelled diagram of Reproductive system of male Rabbit.	2025
17. Explain the difference-between epigeal and hypogeal germination with the help of diagram.	
18. What is fertilization? Describe any five differences between external fertilization and internal fertilization.	
19. What is meant by sexual and asexual reproduction? Describe any five differences.	2024
19. Describe types of asexual reproduction in animals.	2023
16. What is germination? Write the necessary conditions of it and also describe the epigeal and hypogeal germination.	2022
SOME IMPORTANT QUESTIONS FROM NEW BOOK	
Section "B"	
1. Distinguish between asexual and sexual reproduction, Epigeal and Hypogeal Germination.	
2. How a new plant develops with an already growing plant?	
3. How leaves develop into new plants?	
4. Draw neat and labelled diagram of male gametophyte of angiospermic plant?	
5. List out the male reproductive organs of rabbit with glands.	
6. What are STD's?	
7. Why population control is considered important for prosperous society?	
8. Draw life cycle of Angiospermic plant.	
Section "C"	
1. Describe natural asexual reproduction in plants.	
2. What is pollination? Give Adaptive characters of wind and insect pollinated flowers.	
3. Describe the process of Spermatogenesis.	
4. What is germination? Give the condition of germination also describe the methods of different germination.	
UNIT:6	INHERITANCE
Section "B"	
6. Describe variation and explain any two causes of it.	2024
8. Describe any three terms from the following:	

(i) Allele (iii) Homologous chromosomes 10: Describe briefly the structure of DNA.	(ii) Dominant Allele (iv) Recessive Allele	
3. Write three causes of variation. 13. Describe briefly the structure of DNA.		2023
8. Define following terms: (a) Homologous chromosome (b) Allele (c) Recessive Allele 13. State Mendel's law of independent assortment. (No need to draw checkerboard)		2022

SOME IMPORTANT QUESTIONS FROM NEW BOOK

Section "B"

1. What is gene and how it works?
2. Define following terms
(a) Homologous Chromosome b) Heterologous Chromosome
(c) Dominant Allele d) Allele
(e) Recessive Allele f) Homozygous
(g) Heterozygous
3. Which of the phenomena of inheritance is there where both factor express in heterozygous condition.
4. List out the factors which are involved in theory of natural selection.
5. Relate artificial selection with natural selection.

Section "C"

1. Describe the Watson and Crick model of DNA.
2. How gene express describe in detail.
3. State and explain the law of segregation of Mendel's.
4. Explain the inheritance of dihybrid cross by law.
5. Explain the Darwin of theory of natural selection.

UNIT:7

MAN AND HIS ENVIRONMENT

Section "B"

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| 7. Write the names of three mammals and three birds that are extinct endangered in Pakistan. | 2025 |
| 7. What is pollution? How can water pollution be controlled? | 2024 |
| 9. Write note on acid rain. | |
| 9. Write down three methods to control air pollution. | 2023 |
| 4. How does light effect on ecosystem? | 2022 |
| 9. What is algal bloom and how it destroys the life of an aquatic ecosystem? | |

Section "C"

16. Define Ecosystem and describe its biotic factors in detail	2025
19. What is water pollution? How can it be controlled?	
15. Define ecosystem. Describe its biotic components and their types.	2023
16. Define pollution and its types. Write down importance of pollution.	

SOME IMPORTANT QUESTIONS FROM NEW BOOK**Section "B"**

1. Which one is the first trophic level of ecosystem and how?
2. What do we mean by nitrogen fixation and how it occurs in an ecosystem?
3. What is interaction?
4. What is pyramid of number?
5. What is greenhouse effect?
6. What is algal bloom and how it destroys the life of an aquatic ecosystem?
7. What measure can be taken to control water pollution?
8. Write down the name of some endangered mammals of Pakistan.
9. Draw the diagram of nitrogen cycle.

Section "C"

1. Describe biotic factors of an ecosystem.
2. Describe nitrogen cycle as biogeochemical cycle in detail.
3. What is pollution? Describe different types of hazardous effects caused by air pollution.

UNIT:8**BIOTECHNOLOGY****Section "B"**

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| 4. Describe the tools used in Genetic engineering. | 2025 |
| 12. What is fermenter? Write any two advantages of it. | |
| 8. Write down any three uses of fermentation. | 2023 |
| 11. Write down three uses of genetic engineering. | |

Section "C"

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| 17. Define Bio-technology. Explain its role in medicine and agriculture. | 2024 |
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SOME IMPORTANT QUESTIONS FROM NEW BOOK**Section "B"**

1. What kind of enzymes allows scientists to cut and paste pieces of DNA together to form recombinant DNA?

2. Explain how making human tissue plasminogen activator (t-PA) in Chinese hamster ovary (CHO) cells is an example of genetic engineering.
3. How do organisms obtain energy? How does fermentation work?
4. What types of microorganisms cause fermentation to occur?
5. What food and non-food products are created by fermentation?
6. What are some advantages and disadvantages of fermentation in food processing?
7. What factors can affect the fermentation process?
8. Define biotechnology?
9. What is the classical biotechnology?
10. What is your justification in considering biotechnology an old technology?
11. Describe the contribution of Louis Pasteur toward development of classical biotechnology.
12. What are the different types of fermentation?
13. What is lactic acid bacillus? Explain its role in the formation of curd.
14. Name some of the fermented food products of Pakistan.
15. Name four recombinant DNA products available on the market.

UNIT:9	PHARMACOLOGY
Section "B"	
13. How vaccine works against pathogen?	2024
7. What are addictive drugs? Write any three names.	2023
11. Why do antibiotics not affect against viral infection?	2022
Section "C"	
14. What are addictive drugs? Explain any two of them.	2024
SOME IMPORTANT QUESTIONS FROM NEW BOOK	
Section "B"	
1. Why the sedative is used for?	
2. Why addiction is considered as harmful condition?	
3. How drugs (medicine) can be taken from natural source?	
4. Is it possible to get drugs from animals name some of them?	
5. Do we have any harm of antibiotics? If, so mention them.	
6. How vaccine work against pathogen? Explain the process with the help of diagram.	
Section "C"	
1. How bacteria produces antibiotic resistance?	
2. Describe the mode of vaccination.	

DEFINITIONS	Chapter	Years
11. Define the following terms: (i) Myopia	3	2025
(ii) Suckers	5	
(iii) Consumers	7	
2. Define the following: (i) Filtration	2	2023
(ii) Binary Fission	5	
(iii) Evolution	6	

UNIT:	MISCELLANEOUS QUESTIONS
Section "B"	
2. Write three differences between plant cell and animal cell.	2022